

Chapter 2: Literature Review

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LEO Low Earth Orbit
TLE Two Line Elements
JSPOC Joint Space Operations Center
RMS Root Mean Square
EKF Extended Kalman Filter
UKF Unscented Kalman Filter
UPF Unscented Particle Filter
SRIF Square Root Information Filter
GNSS Global Navigation Satellite System
MAGSAT Magnetic Field Satellite
LACE Low-Power Atmospheric Compensation Experiment
DE-2 Dynamics Explorer-2
BLS Batch Least Squares Filter

I. INTRODUCTION

Space has been at the forefront of research and exploration for decades and much of modern technology has come to rely on the innovations and architecture that have been built to maintain it. The largest growing component of space innovation involves the deployment of Low Earth Orbit (LEO) satellites [1]. This greatly expanding font of new technology and ideas brings along its own challenges that range from security to new ways to track orbits [2,3].

II. CUBESATS

A. Security

III. NAVIGATION

LEO satellite use a variety of methods to assist in their own navigation and an organizations ability to track them from the ground. The most common methods used to track a satellite from their ground is through Two Line Elements (TLE)s. TLEs describe the orbital parameters, as seen in Table III, that are required for the tracking and locating of specific satellites during orbit [4]. They provide the specific time and location that each LEO satellite occupies to facilitate communication and to ensure that no satellite impacts another. The recordings of a satellites orbit are taken through observation at Joint Space Operations Center (JSPOC) and provide consistent snapshot updates of currently recorded TLE's every 8 hours [4]. A majority of these satellites have a mean motion of 90 minutes with an altitude(Circular) of 281.49 km [5]. This provides only a few minutes in which the satellite can be contacted before it is out of range and continues until its next orbit. These minimal contact intervals make it vitally important to accurately predict when contact can be made and what

TABLE I
TWO-LINE ELEMENT SET FIELD DEFINITIONS

Line	Field	Content
Title	–	Satellite Name
1	1	Line Number
1	2	Satellite Catalog Number
1	3	Classification (U/C/S)
1	4	International Designator (Launch Year)
1	5	International Designator (Launch Number)
1	6	International Designator (Launch Piece)
1	7	Epoch Year
1	8	Epoch (Day and Fractional Day)
1	9	First Derivative of Mean Motion (rev/day ²)
1	10	Second Derivative of Mean Motion (rev/day ³)
1	11	B* Drag Term (Earth radii ⁻¹)
1	12	Ephemeris Type
1	13	Element Set Number
1	14	Checksum (modulo 10)
2	1	Line Number
2	2	Satellite Catalog Number
2	3	Inclination (deg)
2	4	Right Ascension of Ascending Node (deg)
2	5	Eccentricity
2	6	Argument of Perigee (deg)
2	7	Mean Anomaly (deg)
2	8	Mean Motion (rev/day)
2	9	Revolution Number at Epoch
2	10	Checksum (modulo 10)

communications are of import to send during this window. This creates a need for accurate truly autonomous navigation and orbit determination that does not rely on outside systems to provide accurate information [6].

IV. ALTERNATIVE NAVIGATION

Alternative methods of navigation and orbit determination continue to be explored to provide more robust and accurate systems. A newly explored source of navigation is the use of satellite sensors, like magnetometer and gravimeter, in conjunction with accurate modeling to derive orbit parameters without specific ground station interference [7–9]. These alternative methods provide an independent system of determination without reliance on outside systems [9].

A. Magnetometer

It has been proven by Psiaki and Martel [10] that 5 of the 6 keplerian elements can be determined using only the

TABLE II
POSITION ERROR RESULTS FOR THE SRIF AND BATCH LS FILTERS [6]

S/C	Filter	RMS/Total	Peak	Along/Cross-Track
MAGSAT	SRIF	–	<8 km	slow drift / periodic
MAGSAT	Batch LS	4 km RMS	<8 km	3.4 / 1.8 km RMS
DE-2	SRIF	–	–	not tested
DE-2	Batch LS	–	18 km	8–15 km / steady osc.
LACE	SRIF	–	–	not tested
LACE	Batch LS	88 km RMS	125 km	– / dominant

magnetic field with the missing element being the longitude of the ascending node. Utilizing the magnetometer, known variables like start position and time of flight along with a series of kalman filters the orbit/position of a satellite can be determined [11, 12].

Pisaki and Martel [6] utilize a Square Root Information Filter (SRIF) and Batch Least Squares Filter (BLS) to determine the position of the satellite using only the magnetometer and adjusted based on the truth data. The SRIF was tested on the Magnetic Field Satellite (MAGSAT) launched in October 1979 equipped with a vector magnetometer, the Dynamics Explorer-2 (DE-2) launched in August 1981 equipped with vector magnetometer, and the Low-Power Atmospheric Compensation Experiment (LACE) launched in February 1990 equipped with a scalar magnetometer. The results from the tests can be seen in Table II where the SRIF was able to provide a position estimate with a peak error of less than 8 km for the MAGSAT and the BLS was able to provide a position estimate with a Root Mean Square (RMS) error of 4 km for the MAGSAT, 18 km for the DE-2, and 88 km for the LACE [6]. This demonstrates that the magnetometer can be used to provide a position estimate with a reasonable accuracy, but it is not as accurate as Global Navigation Satellite System (GNSS) and is dependent on the type of magnetometer used.

A series of kalman filters have been explored by Ning and Ma [11] to determine the method and accuracy of differing strategies, this included Unscented Kalman Filter (UKF), Extended Kalman Filter (EKF), and Unscented Particle Filter (UPF). A best case RMS accuracy of 203.32 meters for an EKF, a RMS accuracy of 161.3 meters for an UKF, and a RMS accuracy of 159.7 meters for an UPF when considering a known orbit and 3 second sample rate to filter to a position [11]. The EKF introduces a larger error due to the neglected non-linearities in the system, while the UPF provides a more accurate result at the cost of increased computational complexity [11]. While this method does not provide the same accuracy of GNSS it does provide an independent method of orbit determination that if given the compute power and time can provide a redundant system of orbit determination without reliance on outside systems.

B. Gravitometer

C. ML Models

D. Machine Learning

E. Neural Networks

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